

## 5 Theoretical model

**Notation.** Throughout this section the two springs are numbered 1 (left) and 2 (right). All symbols are collected in Table 2 at the end of the section; each is also defined where it first appears.

The model is built in five steps.

1. Each leaf spring with its magnet is reduced to a mass  $m$  on a spring of stiffness  $k$  (Section 5.1).
2. The repulsive force between the two magnets is written down, treating each magnet as a point magnetic dipole (Section 5.2).
3. The equilibrium of the two springs under this repulsion is described, and the coordinates used for the motion are defined (Section 5.3).
4. The repulsion is *linearised* about the equilibrium separation; this turns the magnetic force into a second spring of stiffness  $\gamma$  joining the two masses (Section 5.4).
5. The resulting pair of coupled equations is solved for the two normal modes, giving the relation that the experiment tests, and for the motion after a single spring is released, which is what is actually recorded (Sections 5.5–5.7).

### 5.1 The leaf spring with its magnet as a mass on a spring

#### 5.1.1 Geometry and symbols

Each leaf spring is a thin rectangular copper strip clamped at its lower end and free at its upper end (Figure 1). The part of the strip above the clamp has *free length*  $L$ , *width*  $b$  (into the page) and *thickness*  $h$  (the small dimension, in the plane of bending). A beam fixed at one end and free at the other is called a *cantilever*. The magnet is glued to the face of the strip at its upper end. When the tip is pushed sideways by a horizontal force  $F$  the strip bends and the tip moves horizontally by a distance  $x$ .

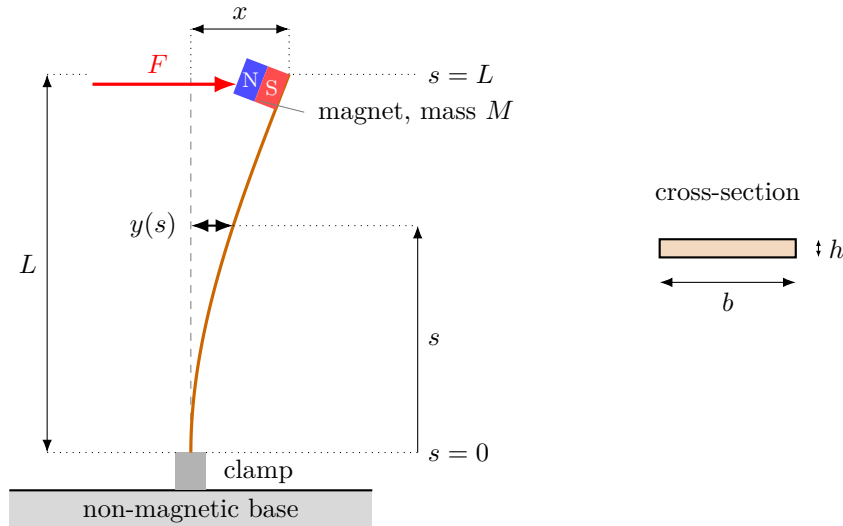


Figure 1: Geometry of the leaf spring.

#### 5.1.2 Why a bent strip pushes back: bending moment and curvature

A straight strip resists being bent because bending stretches the material on the outside of the curve and compresses it on the inside. Consider a short piece of the strip bent into an arc of radius  $R$  (Figure 2). A layer of the strip a distance  $z$  from the middle surface (the *neutral surface*, which is neither stretched nor compressed) has length  $(R + z)\theta$  instead of  $R\theta$ , so its strain (fractional

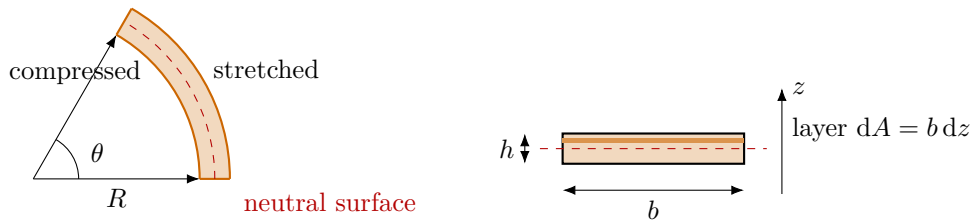
extension) is

$$\varepsilon(z) = \frac{(R+z)\theta - R\theta}{R\theta} = \frac{z}{R}. \quad (1)$$

By Hooke's law for the material, the stress (force per unit area) in that layer is  $\sigma(z) = E\varepsilon(z) = Ez/R$ , where  $E$  is Young's modulus of copper. Tension on one side and compression on the other form a couple: the total turning moment of these internal forces about the neutral surface is

$$\mathcal{M} = \int \sigma(z) z \, dA = \frac{E}{R} \int z^2 \, dA = \frac{EI}{R}, \quad I \equiv \int z^2 \, dA = \int_{-h/2}^{h/2} z^2 b \, dz = \frac{bh^3}{12}. \quad (2)$$

$I$  is called the *second moment of area* of the cross-section; it is a purely geometrical quantity ( $\text{m}^4$ ) that measures how far the material sits from the neutral surface. The thickness enters as  $h^3$ : doubling the thickness makes the strip eight times harder to bend, because the outer layers are both further away (larger strain) and have a longer lever arm.



**Figure 2:** Left: a short element of the strip bent to radius of curvature  $R$  through angle  $\theta$ . Layers outside the neutral surface (dashed) are stretched, layers inside are compressed, by a strain  $z/R$ . Right: the cross-section, with  $z$  measured from the neutral surface; a layer of area  $dA = b \, dz$  at height  $z$  carries stress  $Ez/R$ , and the sum of their moments is  $EI/R$ .

Equation (2) is the *bending equation*: to hold a piece of the strip at curvature  $1/R$  requires a moment  $EI/R$ ; equivalently, wherever a moment  $\mathcal{M}$  acts, the strip curves with  $1/R = \mathcal{M}/EI$ . For small deflections the curvature of the curve  $y(s)$  is its second derivative,  $1/R \approx y''(s)$  (the exact expression is  $y''/(1+y'^2)^{3/2}$ , and  $y'^2 \ll 1$  for the deflections used here), so

$$EI y''(s) = \mathcal{M}(s). \quad (3)$$

### 5.1.3 The static deflection curve of a cantilever with a tip force

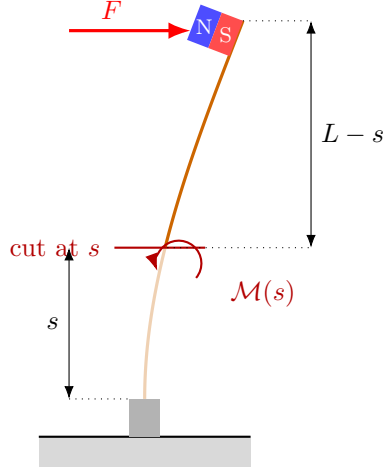
Now apply a sideways force  $F$  at the tip ( $s = L$ ) and nothing else. (Because the strip is vertical, the weights of the strip and magnet act *along* the strip: they produce no bending moment while it is straight, and only a second-order correction once it is bent, which is neglected. This is one advantage of the vertical arrangement.) Cut the strip mentally at a distance  $s$  from the clamp and look at the part above the cut (Figure 3). The only sideways force on that part is  $F$ , acting at the tip, a distance  $L - s$  above the cut. For the part to be in equilibrium the internal moment at the cut must balance the moment of  $F$ :

$$\mathcal{M}(s) = F(L - s). \quad (4)$$

The moment is largest at the clamp ( $s = 0$ ), where the lever arm is the whole length, and zero at the tip: the strip is bent most sharply near the clamp and is straight at the free end, as Figure 1 shows.

Inserting (4) into (3) gives a differential equation for the shape of the strip,

$$EI y''(s) = F(L - s), \quad (5)$$



**Figure 3:** Finding the bending moment. The part of the strip above an imaginary cut at height  $s$  (full colour, together with the magnet) is acted on by the tip force  $F$ , with lever arm  $L - s$ , and by the internal moment  $\mathcal{M}(s)$  exerted on it by the part below the cut (faint). For equilibrium of the upper part,  $\mathcal{M}(s) = F(L - s)$ : largest at the clamp, zero at the tip.

which is solved by integrating twice. The two integration constants are fixed by the clamp, which holds the strip at  $s = 0$  both in position and in direction:  $y(0) = 0$  and  $y'(0) = 0$ .

$$EI y'(s) = F \left( Ls - \frac{s^2}{2} \right) + c_1, \quad y'(0) = 0 \Rightarrow c_1 = 0, \quad (6)$$

$$EI y(s) = F \left( \frac{Ls^2}{2} - \frac{s^3}{6} \right) + c_2, \quad y(0) = 0 \Rightarrow c_2 = 0. \quad (7)$$

Hence

$$y(s) = \frac{F}{6EI} s^2(3L - s), \quad \text{and at the tip} \quad x \equiv y(L) = \frac{F}{6EI} L^2 \cdot 2L = \frac{FL^3}{3EI}. \quad (8)$$

Dividing the shape by the tip deflection gives the *normalised deflection curve*

$$\varphi(s) \equiv \frac{y(s)}{y(L)} = \frac{s^2(3L - s)}{2L^3}, \quad \varphi(0) = 0, \quad \varphi(L) = 1, \quad (9)$$

which says what fraction of the tip's displacement each point of the strip undergoes: zero at the clamp,  $\frac{5}{16} \approx 0.31$  half-way up, and 1 at the tip. (The strip is not bent uniformly: the lower half, where the moment is large, does most of the bending.) This curve is used in Section 5.1.5 to find the effective mass.

#### 5.1.4 The stiffness $k$

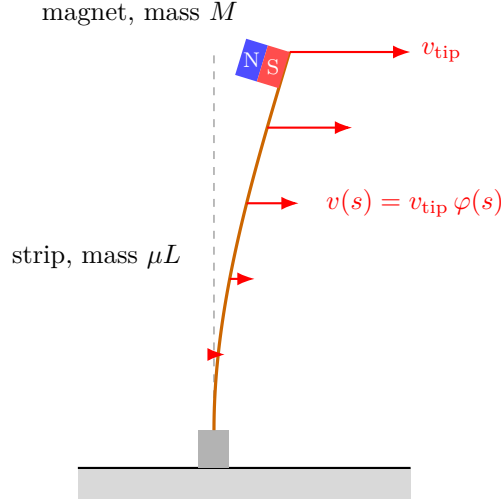
Equation (8) shows that the tip deflection is proportional to the force. The tip therefore behaves as a linear (Hookean) spring with stiffness

$$k \equiv \frac{F}{x} = \frac{3EI}{L^3} = \frac{Ebh^3}{4L^3} \quad (10)$$

Three checks that the formula is sensible: doubling the free length  $L$  makes the spring  $2^3 = 8$  times *softer* (a long strip is floppy); doubling the thickness  $h$  makes it 8 times *stiffer*; doubling the width  $b$  only doubles  $k$  (like two strips side by side).

### 5.1.5 The effective mass $m$

If the strip were massless the oscillating mass would be the magnet alone,  $M$ . But the strip has mass too, and it moves; the difficulty is that it does not all move equally. A point at distance  $s$  from the clamp moves  $\varphi(s)$  times as far as the tip, so during an oscillation it also moves  $\varphi(s)$  times as *fast*: the strip near the clamp is almost stationary while the tip moves with the magnet (Figure 4).



**Figure 4:** Rayleigh's method. During the oscillation every point of the strip is assumed to move in proportion to the static deflection curve  $\varphi(s)$  of equation (9), so its velocity is  $v_{\text{tip}}\varphi(s)$ : nearly zero near the clamp, equal to the magnet's velocity at the tip (arrows). The strip's kinetic energy is the same as that of a point mass  $\frac{33}{140}\mu L$  moving with the tip.

*Rayleigh's method* turns this into a single number by comparing kinetic energies. Let  $\mu$  be the mass per unit length of the strip (so the strip's total mass is  $\mu L$ ). A short piece of strip of length  $ds$  has mass  $\mu ds$  and speed  $v_{\text{tip}}\varphi(s)$ , so the kinetic energy of the whole strip is

$$K_{\text{strip}} = \int_0^L \frac{1}{2} \mu [v_{\text{tip}}\varphi(s)]^2 ds = \frac{1}{2} \left[ \mu \int_0^L \varphi(s)^2 ds \right] v_{\text{tip}}^2. \quad (11)$$

The bracket is the mass of a fictitious point particle that, moving with the tip, would have the same kinetic energy as the strip. With  $\varphi$  from (9), writing  $u = s/L$ ,

$$\int_0^L \varphi^2 ds = \frac{L}{4} \int_0^1 u^4 (3-u)^2 du = \frac{L}{4} \int_0^1 (9u^4 - 6u^5 + u^6) du = \frac{L}{4} \left( \frac{9}{5} - 1 + \frac{1}{7} \right) = \frac{L}{4} \cdot \frac{33}{35} = \frac{33}{140} L. \quad (12)$$

Adding the magnet, the total kinetic energy is  $\frac{1}{2}(M + \frac{33}{140}\mu L)v_{\text{tip}}^2$ , so the system behaves as a single *effective mass*

$$m = M + \frac{33}{140} \mu L \approx M + 0.24 \mu L \quad (13)$$

attached to a spring of stiffness  $k$ . (Strictly,  $M$  should include the glue and any holder, and the magnet's own size adds a small rotational term; both are negligible here.) The assumption that the strip keeps its *static* shape while vibrating is an approximation, but a good one: for a bare strip with no tip mass the exact analysis of the fundamental mode gives a coefficient  $3/3.516^2 = 0.243$  in place of  $33/140 = 0.236$ , a 3% difference in a term that is itself only a fraction of  $m$ ; with a magnet at the tip the true mode shape is even closer to the static tip-load curve. The potential energy of the bent strip is  $\frac{1}{2}kx^2$  by (10), so a single spring with its magnet is a simple harmonic oscillator with

$$\omega_0^2 = \frac{k}{m}, \quad f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}. \quad (14)$$

**Why measure  $k$  and  $m$  instead of calculating them.** Equations (10) and (13) explain how the constants arise, but they are not used to compute them: Young's modulus of a rolled copper strip is uncertain by  $\sim 10\%$ , and  $h^3$  turns a 3% uncertainty in thickness into 9% in  $k$ . Instead, (i) plucking each spring with its own magnet mounted and the other magnet removed gives  $f_0$  and hence  $k/m$  directly, and (ii) a load–deflection graph gives  $k$ ; together they give  $m$ . The test of the exponent in Section 5.6 needs neither, as will be seen.

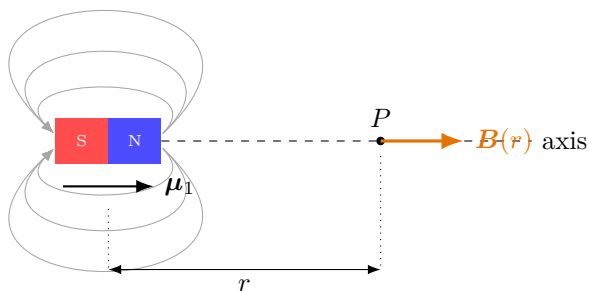
## 5.2 The force between the two magnets

### 5.2.1 A small magnet as a dipole; its field on the axis

A permanent magnet is equivalent, as far as its external field is concerned, to a small loop of current. Its strength is measured by its *magnetic dipole moment*  $\mu_m$  (unit  $\text{A m}^2$ ; for a loop of area  $S$  carrying current  $I$ ,  $\mu_m = IS$ ). The field of a dipole at distances large compared with its size has a standard form; on the dipole's own axis, at distance  $r$  from its centre, the field is directed along the axis and has magnitude

$$B(r) = \frac{\mu_0}{4\pi} \frac{2\mu_m}{r^3}, \quad (15)$$

where  $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$  is the permeability of free space. (Equation (15) follows from the Biot–Savart field on the axis of a current loop of radius  $a$ ,  $B = \mu_0 I a^2 / [2(a^2 + r^2)^{3/2}]$ , in the limit  $r \gg a$ :  $B \rightarrow \mu_0 I a^2 / 2r^3 = \mu_0 (I\pi a^2) / 2\pi r^3 = \mu_0 2\mu_m / 4\pi r^3$ .) The essential feature is the *inverse cube*. For a real magnet,  $\mu_m \approx B_r V / \mu_0$ , where  $B_r$  is the remanence quoted by the manufacturer and  $V$  the magnet's volume.



**Figure 5:** The field of a small magnet (a magnetic dipole of moment  $\mu_1$ , magnitude  $\mu_m$ , pointing from S to N). Grey: field lines. At a point  $P$  on the axis, at distance  $r$  from the *centre* of the magnet, the field points along the axis in the direction of the moment, with magnitude  $\mu_0 2\mu_m / 4\pi r^3$ .

### 5.2.2 Energy and force of the second magnet

Place a second identical magnet on the same axis with its like pole facing the first (Figure 6). A dipole  $\mu_2$  in a field  $B$  has potential energy

$$U = -\mu_2 \cdot B, \quad (16)$$

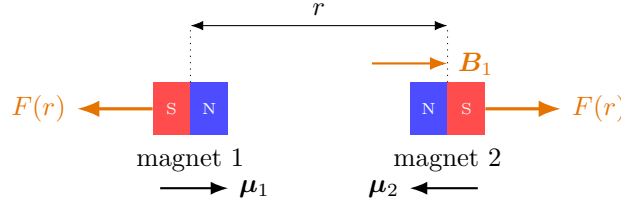
lowest when the moment is parallel to the field (a compass needle aligns with the field). With like poles facing,  $\mu_2$  points *against*  $B_1$ , so the energy is positive:

$$U(r) = +\mu_m B_1(r) = \frac{\mu_0}{4\pi} \frac{2\mu_m^2}{r^3}. \quad (17)$$

The force on magnet 2 along the axis is minus the derivative of this energy with respect to  $r$ :

$$F(r) = -\frac{dU}{dr} = \frac{\mu_0}{4\pi} \frac{6\mu_m^2}{r^4} = \frac{3\mu_0\mu_m^2}{2\pi r^4} \quad (18)$$

The sign is positive, i.e. the force tends to *increase*  $r$ : the magnets repel, and by Newton's third law magnet 1 feels an equal force the other way. The force falls as the *fourth* power of the separation because the field falls as  $r^{-3}$  and the force is proportional to its gradient.



**Figure 6:** Two identical magnets on a common axis with like poles (N) facing. The field  $\mathbf{B}_1$  of magnet 1 at magnet 2 points to the right; the moment  $\boldsymbol{\mu}_2$  of magnet 2 points to the left, so the energy  $-\boldsymbol{\mu}_2 \cdot \mathbf{B}_1$  is positive and decreases with  $r$ : the magnets repel with  $F(r) = 3\mu_0\mu_m^2/2\pi r^4$ , equal and opposite on the two magnets.

**Two remarks on  $F(r)$ .** (i) The  $r$  in (18) is the distance between the *centres* of the magnets, not the gap between their faces; this is why the separation  $d$  is defined centre-to-centre throughout the essay ( $d = \text{gap} + \text{thickness}$  for two equal magnets). (ii) The dipole formula is exact only in the limit  $r \gg D$  ( $D = \text{magnet diameter}$ ). Here  $r/D$  is only of order 2–6, so deviations are expected at the smallest separations; this is assumption (a) below and is examined in the evaluation.

### 5.3 Equilibrium and coordinates

With both magnets mounted the repulsion pushes the two tips apart until each spring's restoring force balances it (Figure 7). If the tips end up a distance  $d$  apart (centre to centre), each spring is bent outwards by a static deflection  $\delta$  obeying

$$k\delta = F(d). \quad (19)$$

If the clamps were set a distance  $d_0$  apart, then  $d = d_0 + 2\delta$ ; finding  $d$  from  $d_0$  would require solving  $k(d - d_0)/2 = 3\mu_0\mu_m^2/2\pi d^4$ , a fifth-degree equation containing two constants that are not accurately known. The essay avoids this by *measuring*  $d$  directly in the equilibrium state, with both magnets mounted and the springs already bent.

From here on:

- $x_1$  and  $x_2$  are the horizontal displacements of magnets 1 and 2 from their *equilibrium* positions, both counted positive in the same direction (to the right in Figure 7);
- the instantaneous centre-to-centre separation is  $r = d + x_2 - x_1$  (if magnet 2 moves right and magnet 1 stays, the gap grows);
- the *unstressed* (straight) position of spring 1 lies  $\delta$  to the right of its equilibrium (spring 1 is bent to the left), and that of spring 2 lies  $\delta$  to the left of its equilibrium. This is needed for the spring forces in Section 5.4.

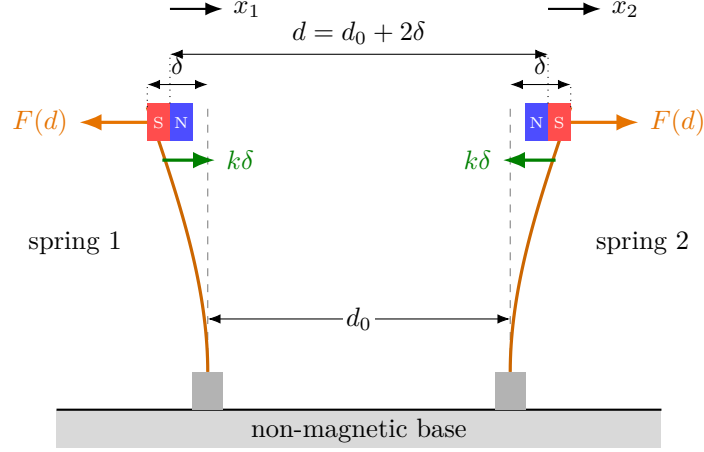
### 5.4 Linearising the repulsion: the magnetic “spring” $\gamma$

During the motion the separation is  $r = d + u$  with  $u = x_2 - x_1$  the change of separation. Since the amplitudes are small compared with  $d$ , the force can be replaced by the first two terms of its Taylor series about  $r = d$  (geometrically: the curve  $F(r)$  is replaced by its tangent at  $r = d$ , Figure 8):

$$F(d + u) \approx F(d) + F'(d)u \quad \Longleftrightarrow \quad F(d + u) - F(d) \approx -\gamma u, \quad \gamma \equiv -F'(d). \quad (20)$$

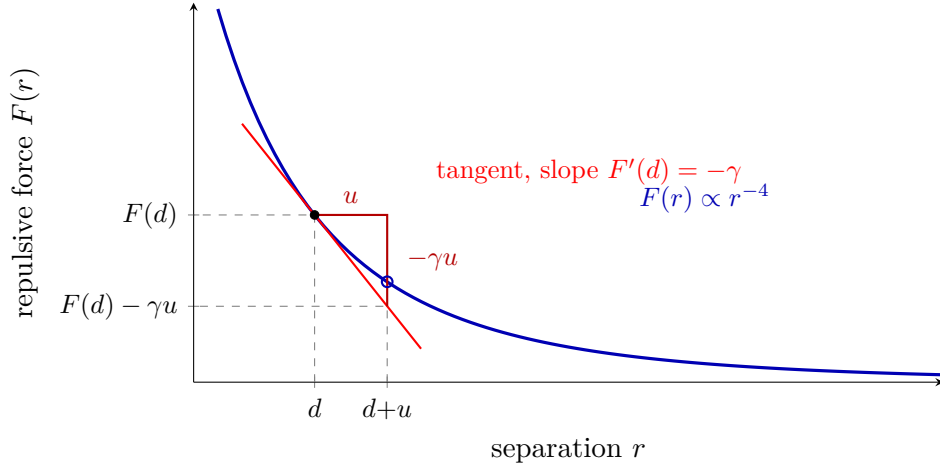
Differentiating (18) with  $d(r^{-4})/dr = -4r^{-5}$ ,

$$\boxed{\gamma = -\left.\frac{dF}{dr}\right|_{r=d} = \frac{4F(d)}{d} = \frac{6\mu_0\mu_m^2}{\pi d^5}} \quad (21)$$



**Figure 7:** Equilibrium with both magnets mounted. The repulsion  $F(d)$  bends each spring outwards by  $\delta$  from its straight (dashed) position until the elastic restoring force  $k\delta$  balances it. The centre-to-centre separation  $d$  is defined and measured in this state ( $d_0$  is the spacing of the clamps). The displacements  $x_1$ ,  $x_2$  used in the equations of motion are measured from these equilibrium positions, both positive to the right.

$\gamma$  is the *magnetic stiffness*: the change in the repulsive force per unit change of separation. It is positive because a repulsion that weakens with distance always pushes the magnets back towards their equilibrium separation—if they come closer the push apart grows, if they move apart it weakens and the springs win. So, for small motions, the magnetic field between the magnets behaves exactly like a compressed spring of stiffness  $\gamma$  connecting them. This is the crucial step for the exponent: the force goes as  $d^{-4}$ , but the frequencies will turn out to depend on the *stiffness*, which is the derivative of the force and goes as  $d^{-5}$ .



**Figure 8:** Linearisation of the force law. Near the equilibrium separation  $d$  the curve  $F(r)$  is replaced by its tangent, whose slope is  $-\gamma$ . Increasing the separation by  $u$  reduces the repulsion by  $\gamma u$ ; decreasing it increases the repulsion by the same amount—the behaviour of a compressed spring of stiffness  $\gamma$ . The gap between the tangent and the true curve (open circle) is the error of the approximation; it grows with  $u/d$  and is discussed in the evaluation.

## 5.5 Equations of motion and normal modes

### 5.5.1 Newton's second law for each magnet

Figure 9 shows the horizontal forces on each magnet when both are displaced. Consider magnet 1. Its displacement from the *unstressed* position of spring 1 is  $x_1 - \delta$  (Section 5.3), so the spring

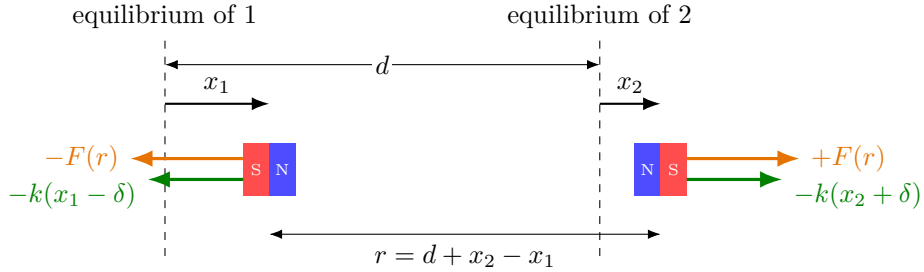
exerts  $-k(x_1 - \delta)$ . The magnetic force on it is  $-F(r)$ : it is pushed away from magnet 2, i.e. in the negative direction, with the magnitude evaluated at the current separation  $r = d + u$ . Newton's second law, with the effective mass  $m$  of (13):

$$\begin{aligned} m\ddot{x}_1 &= -k(x_1 - \delta) - F(d + u) \\ &= -kx_1 + \underbrace{k\delta}_{=F(d) \text{ by (19)}} - F(d + u) = -kx_1 + [F(d) - F(d + u)] \approx -kx_1 + \gamma u, \end{aligned} \quad (22)$$

where the last step used (20). Two things happened here. The constant part of the magnetic force,  $F(d)$ , was exactly cancelled by the static bend  $k\delta$ : that is what “measured from equilibrium” buys. Only the *change* of the magnetic force survives, and it is proportional to the change of separation  $u = x_2 - x_1$ .

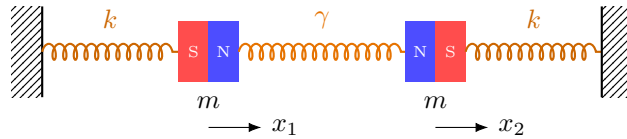
Magnet 2 is the mirror image: its spring force is  $-k(x_2 + \delta)$  (its unstressed position is  $\delta$  to the *left*), the magnetic force on it is  $+F(d + u)$ , and  $k\delta = F(d)$  again cancels the constant part, leaving  $m\ddot{x}_2 = -kx_2 - \gamma u$ . Together:

$$\boxed{m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1)} \quad (23)$$



**Figure 9:** Free-body diagram of the two magnets (horizontal forces only, seen from above; orange: magnetic, green: elastic), both shown displaced to the right of their equilibrium positions. Each spring force is proportional to the displacement from that spring's *unstressed* position ( $+\delta$  for spring 1,  $-\delta$  for spring 2, Figure 7). The magnetic force has magnitude  $F(r)$  at the *current* separation and pushes the magnets apart.

Equations (23) are exactly those of two equal masses on equal springs  $k$  joined by a third spring of stiffness  $\gamma$  (Figure 10): the middle spring exerts  $+\gamma(x_2 - x_1)$  on the left mass and  $-\gamma(x_2 - x_1)$  on the right one. The magnetic field has become that third spring.



**Figure 10:** Mechanical equivalent of equations (23): two equal masses  $m$  on equal springs  $k$  (copper: the leaf springs), coupled by a spring of stiffness  $\gamma$  (orange: the magnetic repulsion). The picture is schematic; in the apparatus the coupling “spring” is the field between the magnets.

**A common slip.** The magnetic term is  $\gamma(x_2 - x_1)$ , not  $-\gamma x_1$ . The repulsion depends on the *separation* of the magnets, not on where magnet 1 happens to be: if both magnets move together the separation does not change and the magnetic force does not change at all.

### 5.5.2 Decoupling: the normal coordinates

The two equations (23) are coupled (each contains both  $x_1$  and  $x_2$ ), but they can be separated by adding and subtracting them. Adding, the coupling terms cancel:

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2) \implies m\ddot{\eta} = -k\eta, \quad \eta \equiv x_1 + x_2. \quad (24)$$



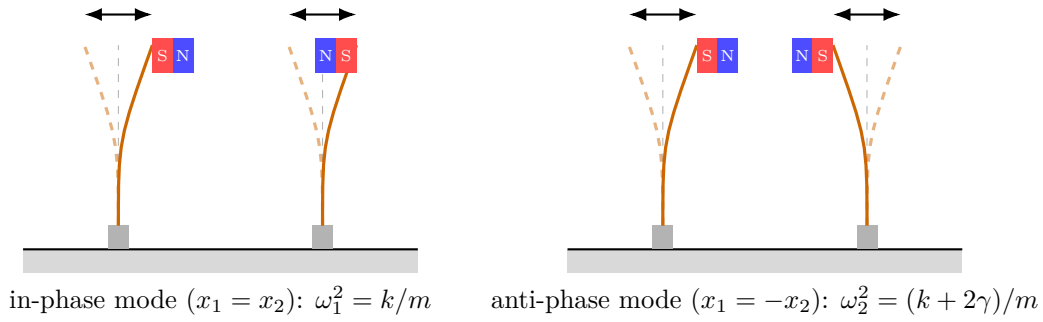
Subtracting, the coupling terms add:

$$m(\ddot{x}_1 - \ddot{x}_2) = -k(x_1 - x_2) + 2\gamma(x_2 - x_1) = -(k + 2\gamma)(x_1 - x_2) \implies m\ddot{\xi} = -(k + 2\gamma)\xi, \quad \xi \equiv x_1 - x_2. \quad (25)$$

$\eta$  and  $\xi$  are the *normal coordinates*. Each obeys the simple-harmonic equation  $\ddot{q} = -\omega^2 q$  on its own, with

$$\omega_1^2 = \frac{k}{m} \quad (\eta, \text{ in-phase}), \quad \omega_2^2 = \frac{k + 2\gamma}{m} \quad (\xi, \text{ anti-phase}) \quad (26)$$

A motion in which only  $\eta$  is non-zero has  $\xi = 0$ , i.e.  $x_1 = x_2$ : the two magnets move together, keeping their separation, at angular frequency  $\omega_1$  (Figure 11, left). A motion with only  $\xi$  non-zero has  $x_1 = -x_2$ : the magnets move in opposite directions, alternately approaching and receding, at  $\omega_2$  (right). These two patterns are the *normal modes*: if the system is started exactly in one of them it stays in it, oscillating at a single frequency. Every other motion is a mixture of the two (Section 5.7).



**Figure 11:** The two normal modes (solid: one extreme of the swing; dashed: the other; magnets drawn at the solid extreme). Left, in-phase: the magnets keep their separation, the magnetic force never changes from its equilibrium value, and each spring oscillates as if alone, at  $\omega_1 = \sqrt{k/m}$ . Right, anti-phase: when each magnet moves  $a$  towards the other the separation shrinks by  $2a$ , so each magnet feels an extra restoring force  $\gamma \cdot 2a$  on top of  $ka$ ; the effective stiffness is  $k + 2\gamma$ .

The factor 2 in  $\omega_2^2$  has a simple origin, visible in Figure 11: in the anti-phase mode a displacement  $a$  of each magnet changes the separation by  $2a$ , so the magnetic spring, which responds to the separation, exerts  $2\gamma a$  on each magnet while the magnet itself has moved only  $a$ . Equivalently, in matrix form  $m\ddot{\mathbf{x}} = -\mathbf{K}\mathbf{x}$  with  $\mathbf{K} = \begin{pmatrix} k + \gamma & -\gamma \\ -\gamma & k + \gamma \end{pmatrix}$ , whose eigenvalues are  $k$  and  $k + 2\gamma$  with eigenvectors  $(1, 1)$  and  $(1, -1)$ .

Two limits confirm the result. As  $d \rightarrow \infty$ ,  $\gamma \rightarrow 0$  and both frequencies become  $\sqrt{k/m}$ : two uncoupled springs. As  $d$  decreases,  $\gamma$  grows as  $d^{-5}$ ,  $\omega_2$  rises steeply while  $\omega_1$  does not move. The size of the coupling can be read from the data:  $2\gamma/k = \omega_2^2/\omega_1^2 - 1 = (f_2/f_1)^2 - 1$ , which is 0.67 at  $d = 21$  mm (the magnetic spring is about one third as stiff as a leaf spring) and 0.02 at 40 mm.

## 5.6 The relation tested in this essay

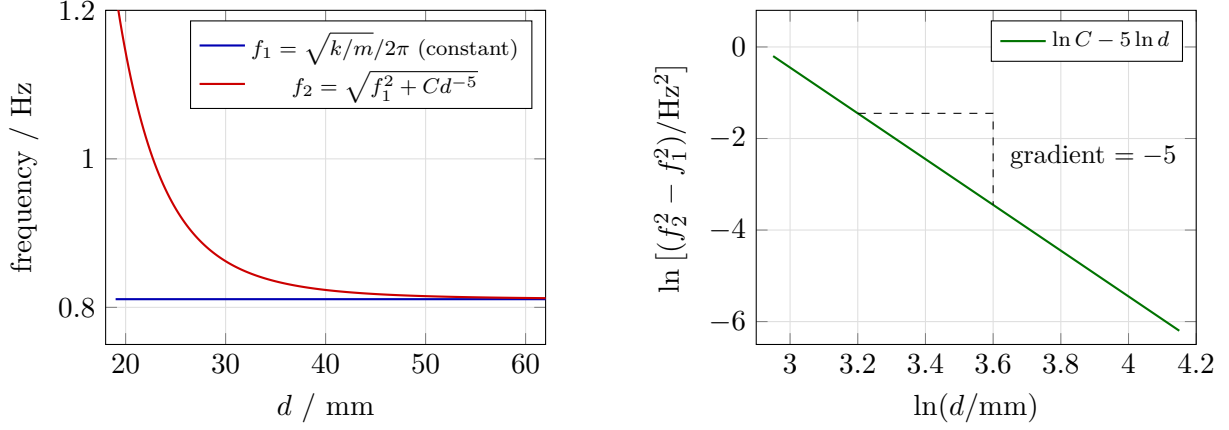
With  $\omega = 2\pi f$ , subtracting the two results of (26) eliminates  $k$  entirely:

$$f_2^2 - f_1^2 = \frac{\omega_2^2 - \omega_1^2}{4\pi^2} = \frac{2\gamma}{4\pi^2 m} = \frac{\gamma}{2\pi^2 m} = \frac{3\mu_0\mu_m^2}{\pi^3 m} d^{-5} \equiv C d^{-5} \quad (27)$$

(the last step inserts  $\gamma$  from (21):  $6/(2\pi^2 \cdot \pi) = 3/\pi^3$ ). Taking natural logarithms,

$$\ln(f_2^2 - f_1^2) = \ln C - 5 \ln d, \quad (28)$$

so a graph of  $\ln(f_2^2 - f_1^2)$  against  $\ln d$  should be a straight line of gradient  $-5$  and intercept  $\ln C$ . The gradient tests the *form* of the force law ( $F \propto r^{-4} \implies$  gradient  $-5$ ); the intercept tests its *magnitude*, and requires  $\mu_m$  and  $m$ .



**Figure 12:** The prediction of the model, drawn with  $f_1 = 0.811$  Hz and the value of  $C$  obtained from the data. Left:  $f_1$  is independent of  $d$  while  $f_2$  falls towards it. Right: the coupling term  $f_2^2 - f_1^2$  on logarithmic axes is a straight line of gradient exactly  $-5$ .

**Why  $f_2^2 - f_1^2$  and not simply  $f_2$ ?** (i)  $k$  cancels, so the spring constant, Young's modulus and the strip dimensions are not needed to test the force law. (ii)  $f_1$  is measured in the *same* record as  $f_2$ , so a slow drift of the springs (temperature, clamp creep) affects both and largely cancels. (iii) The result is linear in  $\gamma$ , so the exponent of  $d$  appears directly as the gradient of a straight line.

**Units.**  $[\mu_0 \mu_m^2] = \text{T m A}^{-1} \cdot \text{A}^2 \text{m}^4 = \text{T A m}^5$  and  $1 \text{ T} = 1 \text{ kg A}^{-1} \text{s}^{-2}$ , so  $[\mu_0 \mu_m^2] = \text{kg m}^5 \text{s}^{-2}$ . Dividing by  $m$  (kg) and  $d^5$  ( $\text{m}^5$ ) leaves  $\text{s}^{-2}$ , the units of  $f^2$ .  $C$  has units  $\text{m}^5 \text{s}^{-2}$ .

**Static deflection.** From (19), (21) and (27),  $\delta = F(d)/k = \gamma d/4k = \pi^2(f_2^2 - f_1^2)d/2\omega_1^2$ : about 1.8 mm at 21 mm and 0.1 mm at 40 mm—small compared with  $d$ , but comparable to the uncertainty in  $d$ , which is one more reason to measure  $d$  at equilibrium.

## 5.7 Motion after releasing one spring: beats

### 5.7.1 The general motion and the mode amplitudes

Because  $\eta$  and  $\xi$  each perform simple harmonic motion, the most general motion is

$$\eta(t) = \eta_0 \cos(\omega_1 t + \phi_1), \quad \xi(t) = \xi_0 \cos(\omega_2 t + \phi_2), \quad (29)$$

with four constants fixed by the four initial conditions (two positions, two velocities). The frequencies are properties of the system; the *amounts*  $\eta_0$ ,  $\xi_0$  of each mode depend only on how the motion is started. In the experiment both springs are released from rest,  $\dot{x}_1(0) = \dot{x}_2(0) = 0$ , hence  $\dot{\eta}(0) = \dot{\xi}(0) = 0$ , which forces  $\phi_1 = \phi_2 = 0$ . The mode amplitudes are then simply the initial values of the normal coordinates:

$$\eta_0 = x_1(0) + x_2(0), \quad \xi_0 = x_1(0) - x_2(0). \quad (30)$$

Transforming back with  $x_1 = (\eta + \xi)/2$  and  $x_2 = (\eta - \xi)/2$ :

$$x_1(t) = \frac{\eta_0}{2} \cos \omega_1 t + \frac{\xi_0}{2} \cos \omega_2 t, \quad x_2(t) = \frac{\eta_0}{2} \cos \omega_1 t - \frac{\xi_0}{2} \cos \omega_2 t \quad (31)$$

Each spring's displacement is the sum of *two cosines*, one at each mode frequency. Table 1 lists what different starting conditions produce.

### 5.7.2 What the Fourier spectrum shows, and why the two peaks are not equal

A Fourier transform of a record reports the amplitude of the cosine present at each frequency. By (31) the record  $x_1(t)$  contains exactly two cosines, so its spectrum shows two peaks, at  $f_1$  and

**Table 1:** Mode amplitudes for different ways of starting the motion (both springs released from rest).

| how the motion is started                                      | $x_1(0), x_2(0)$ | $\eta_0, \xi_0$          | resulting motion                                                    |
|----------------------------------------------------------------|------------------|--------------------------|---------------------------------------------------------------------|
| both displaced the same way                                    | $A, A$           | $2A, 0$                  | pure in-phase mode: single frequency $f_1$                          |
| displaced in opposite directions                               | $A, -A$          | $0, 2A$                  | pure anti-phase mode: single frequency $f_2$                        |
| spring 1 displaced, spring 2 exactly at equilibrium            | $A, 0$           | $A, A$                   | both modes, equal amplitude: beats whose minima reach zero          |
| spring 1 <i>held</i> at $A$ ; spring 2 has shifted to $x_{20}$ | $A, x_{20}$      | $A + x_{20}, A - x_{20}$ | both modes, unequal amplitude: beats whose minima do not reach zero |

$f_2$ , and (for a record long enough to separate them) their heights are proportional to  $\eta_0/2$  and  $\xi_0/2$ :

$$\frac{\text{height of the peak at } f_2}{\text{height of the peak at } f_1} = \frac{\xi_0}{\eta_0}. \quad (32)$$

Two consequences are important for the method. First, the *positions* of the peaks,  $f_1$  and  $f_2$ , do not depend on the starting condition at all: a single FFT of a single spring gives both mode frequencies however the spring was released, provided both modes are present. Second, the *heights* depend only on the starting condition, and are equal only in the idealised case  $x_2(0) = 0$ .

That ideal case is not what happens in practice. Spring 1 is *held* at  $x_1 = A$  for a moment before it is released, and while it is held the repulsion on spring 2 differs from its equilibrium value, so spring 2 shifts to a new rest position before the release. Setting  $\ddot{x}_2 = 0$  in the second of (23) with  $x_1 = A$ :

$$0 = -kx_{20} - \gamma(x_{20} - A) \implies x_{20} = \frac{\gamma}{k + \gamma} A, \quad (33)$$

in the same direction as  $A$  (spring 2 is pushed away if spring 1 is pushed towards it, and follows if spring 1 is pulled away). By (30) and (32),

$$\boxed{\frac{\text{height at } f_2}{\text{height at } f_1} = \frac{A - x_{20}}{A + x_{20}} = \frac{k}{k + 2\gamma} = \frac{\omega_1^2}{\omega_2^2} = \left(\frac{f_1}{f_2}\right)^2} \quad (34)$$

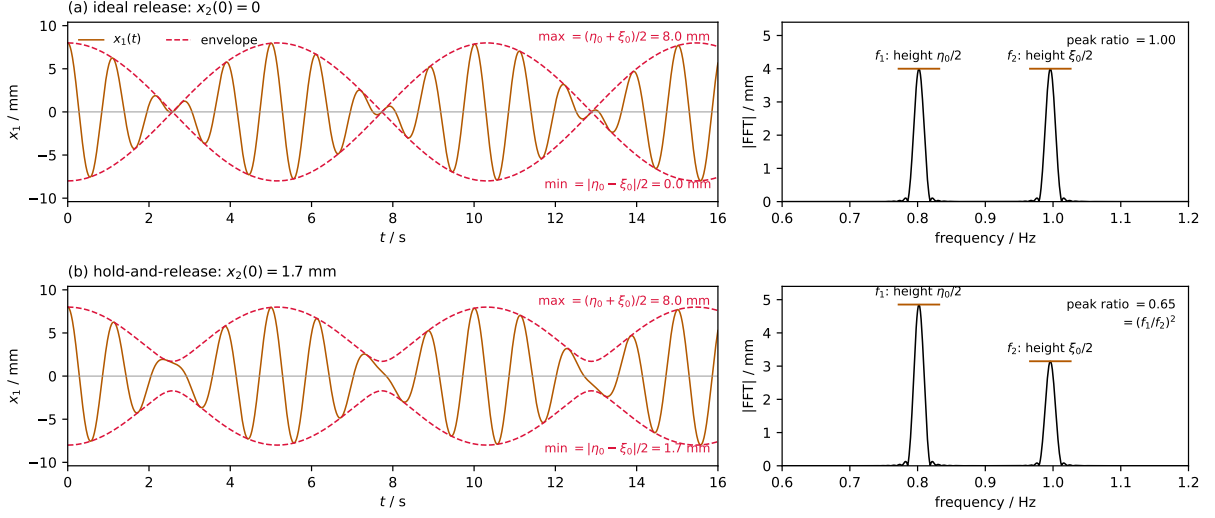
With the frequencies measured at  $d = 23$  mm ( $f_1 = 0.802$ ,  $f_2 = 0.996$  Hz) this predicts a ratio of 0.65; the observed ratio is about 0.6. The residual difference is of the size expected from the non-linearity of the force at  $A = 10$  mm (evaluation). The same shift explains why the beat minima are not zero: the amplitude of  $x_1$  swings between  $(\eta_0 + \xi_0)/2 = A$  and  $(\eta_0 - \xi_0)/2 = x_{20}$ , so minimum/maximum  $= \gamma/(k + \gamma) \approx 0.2$  (Figure 13). Neither effect moves  $f_1$  or  $f_2$ .

### 5.7.3 Why two cosines make beats

Represent each cosine as the horizontal projection of a rotating arrow (Figure 14): arrow 1 of length  $\eta_0/2$  turning at  $f_1$  revolutions per second, arrow 2 of length  $\xi_0/2$  turning at  $f_2$  revolutions per second;  $x_1$  is the horizontal projection of their vector sum. Since arrow 2 turns slightly faster, the angle between them grows steadily, by  $f_2 - f_1$  revolutions per second. When the arrows are aligned the sum is long,  $(\eta_0 + \xi_0)/2$ ; half a relative revolution later they oppose and the sum is short,  $|\eta_0 - \xi_0|/2$ ; after one full relative revolution they are aligned again. So the *amplitude* of the fast oscillation swells and shrinks with period

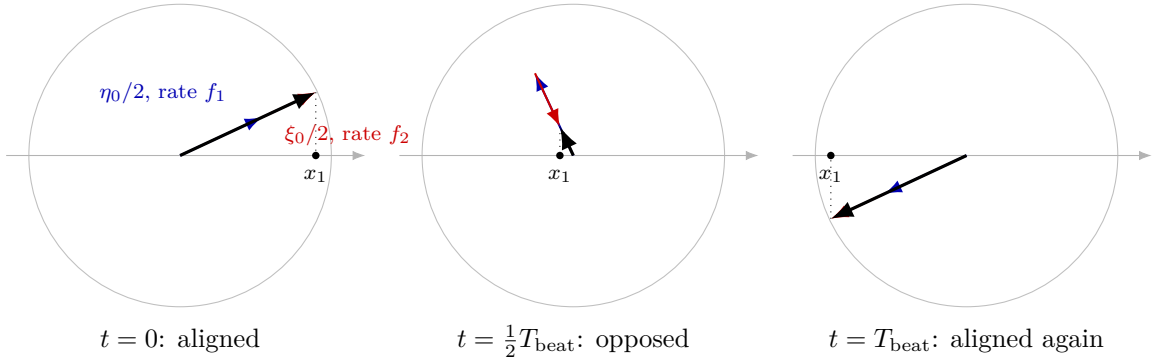
$$\boxed{T_{\text{beat}} = \frac{1}{f_2 - f_1}, \quad f_{\text{beat}} \equiv \frac{1}{T_{\text{beat}}} = f_2 - f_1} \quad (35)$$

Same  $f_1 = 0.802$  Hz,  $f_2 = 0.996$  Hz in both rows; only the starting condition differs (linear model,  $A = 8$  mm)



**Figure 13:** The same system ( $f_1 = 0.802$ ,  $f_2 = 0.996$  Hz; linear model,  $A = 8$  mm) started in two ways; left:  $x_1(t)$  with its envelope, right: its amplitude spectrum, with the predicted peak heights  $\eta_0/2$  and  $\xi_0/2$  marked. Top: idealised instantaneous release with  $x_2(0) = 0$ —equal mode amplitudes, equal FFT peaks, beat minima at zero. Bottom: hold-and-release,  $x_2(0) = A\gamma/(k+\gamma) = 1.7$  mm—peak ratio  $(f_1/f_2)^2 = 0.65$  and beat minima of 1.7 mm. The peak *positions* are identical in both cases. (Generated by `make_fig_release.py`.)

whatever the two lengths are; the lengths only decide how deep the minima go. Meanwhile the sum-arrow itself turns at approximately the mean rate, so the fast oscillation has frequency close to  $f_{\text{avg}} = (f_1 + f_2)/2$  (exactly so for equal lengths).



**Figure 14:** Beats as rotating arrows (phasors). Blue: length  $\eta_0/2$ , turning at  $f_1$  revolutions per second. Red, drawn head-to-tail: length  $\xi_0/2$ , turning at  $f_2$ . Black: their sum;  $x_1$  is its horizontal projection. The relative angle grows by one full turn every  $1/(f_2 - f_1)$  seconds, so the length of the black arrow—the amplitude of the fast oscillation—swings between  $(\eta_0 + \xi_0)/2$  and  $|\eta_0 - \xi_0|/2$  with exactly that period. Drawn for  $\xi_0/\eta_0 = 0.65$ , the hold-and-release case: the amplitude never falls to zero.

For the equal-amplitude case  $\eta_0 = \xi_0 = A$  the same result follows from the identity  $\cos \alpha + \cos \beta = 2 \cos \frac{\alpha-\beta}{2} \cos \frac{\alpha+\beta}{2}$  with  $\alpha = 2\pi f_1 t$ ,  $\beta = 2\pi f_2 t$ :

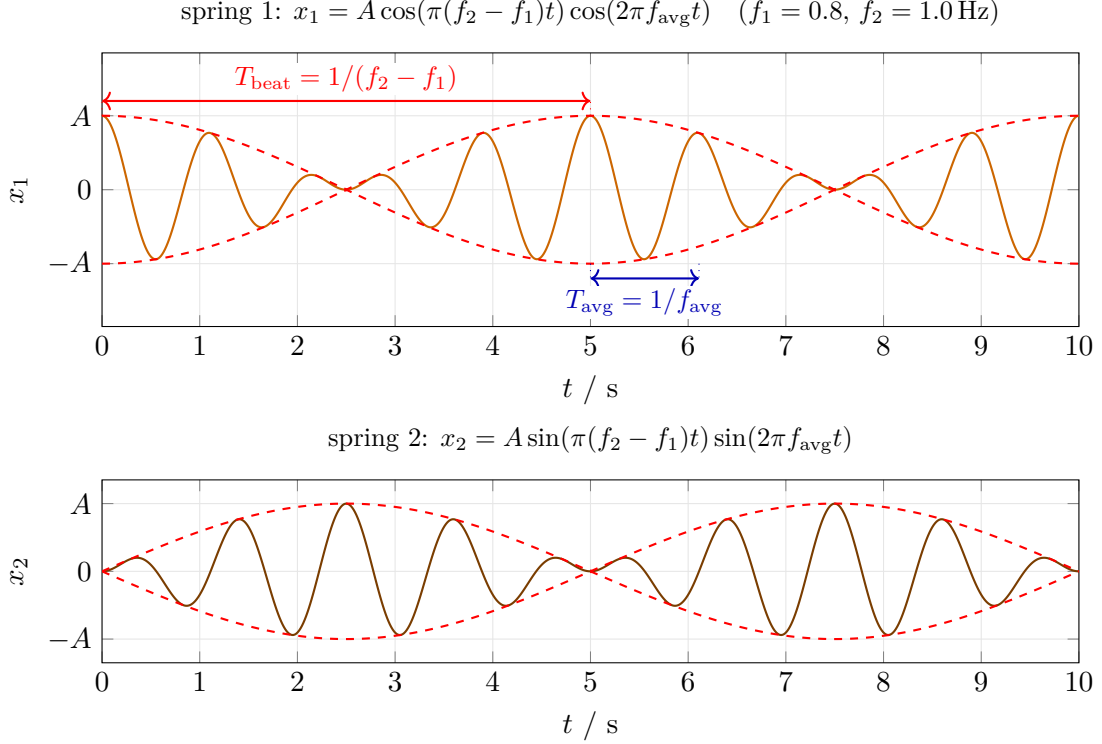
$$x_1(t) = A \cos(\pi(f_2 - f_1)t) \cos(2\pi f_{\text{avg}} t), \quad f_{\text{avg}} = \frac{f_1 + f_2}{2} \quad (36)$$

a fast oscillation at the average frequency whose amplitude  $A \cos(\pi(f_2 - f_1)t)$  varies slowly (Figure 15). The envelope cosine has frequency  $(f_2 - f_1)/2$ , but what is observed is its *modulus*, which

has two maxima per cycle; the swellings therefore recur every  $1/(f_2 - f_1)$ , in agreement with (35). Multiplying out  $(f_2 - f_1)(f_2 + f_1)$  gives the link to the coupling term:

$$\boxed{f_2^2 - f_1^2 = 2 f_{\text{beat}} f_{\text{avg}}} \quad (37)$$

The beat frequency is the mode splitting itself, and the quantity tested in (27) is the product of the splitting and twice the average frequency.



**Figure 15:** Beats for equal mode amplitudes. Top: the released spring and its envelope (red dashed); the swellings recur every  $T_{\text{beat}} = 1/(f_2 - f_1)$ , and the fast oscillation has period  $1/f_{\text{avg}}$ . Bottom: spring 2 carries the same two frequencies with the anti-phase term reversed,  $x_2 = \frac{A}{2}(\cos \omega_1 t - \cos \omega_2 t) = A \sin(\pi(f_2 - f_1)t) \sin(2\pi f_{\text{avg}}t)$ ; its envelope is a quarter of a beat behind, so it swings most when spring 1 is momentarily still. This is the energy transfer between the springs.

#### 5.7.4 Spring 2, and a test of identical springs

Spring 2 contains the same two cosines with the sign of the anti-phase term reversed (equation (31)). Hence (i) its spectrum has peaks at the same  $f_1$  and  $f_2$ ; (ii) for identical springs the height ratio  $\xi_0/\eta_0$  is the *same* for both springs whatever the starting condition, because each mode has equal amplitude in both springs (mode shapes (1, 1) and (1, -1)); (iii) if the springs are not identical the mode shapes tilt and the two ratios differ, so comparing them is a direct test of the identical-spring assumption (used in the evaluation).

#### 5.7.5 Predictions for the average and beat frequencies

Combining  $f_2 = \sqrt{f_1^2 + Cd^{-5}}$  from (27) with the definitions,

$$f_{\text{beat}} = \sqrt{f_1^2 + Cd^{-5}} - f_1 = \frac{Cd^{-5}}{2f_{\text{avg}}}, \quad f_{\text{avg}} = \frac{f_1 + \sqrt{f_1^2 + Cd^{-5}}}{2}. \quad (38)$$

Because  $f_{\text{avg}}$  can only vary between  $f_1$  (weak coupling) and about  $1.15f_1$  (closest separation used), while  $d^{-5}$  varies by a factor of 25 over the range, the beat frequency falls almost as steeply as

$d^{-5}$  itself—for weak coupling  $f_{\text{beat}} \approx Cd^{-5}/2f_1$  exactly follows the inverse-fifth-power law—while the average frequency is nearly constant and rises above  $f_1$  only where the coupling is strong. Numerically, the beat period should lengthen from about 4 s at 21 mm to over 100 s at 40 mm and to some ten minutes at 60 mm; this is why no second peak can be resolved at the largest separations in a 30 s record.

## 5.8 Assumptions

The derivation assumed:

- (a) the magnets are point dipoles ( $d \gg D$ ) — entered in (18);
- (b) the displacements are small enough for the force to be linearised ( $A \ll d$ ) — entered in (20);
- (c) the two springs are identical ( $k$  and  $m$  the same for both) — needed for the decoupling in (26);
- (d) each spring can be represented by a single lumped mass — (13);
- (e) there is no damping;
- (f) the magnets stay coaxial (the tip of a bending cantilever actually tilts slightly) — (15);
- (g) the FFT returns the true frequencies of the motion.

Each is revisited in the evaluation, where its sign, size and remedy are estimated.

**Table 2:** Symbols used in this section.

| Symbol                        | Meaning                                                            | Symbol                                             | Meaning                                                                    |
|-------------------------------|--------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------|
| $L, b, h$                     | free length, width, thickness of a strip                           | $d$                                                | equilibrium centre-to-centre separation of the magnets                     |
| $s, y(s)$                     | distance along the strip from the clamp; sideways deflection there | $d_0$                                              | spacing of the clamps ( $d = d_0 + 2\delta$ )                              |
| $E$                           | Young’s modulus of copper                                          | $\delta$                                           | static outward bend of each spring, $k\delta = F(d)$                       |
| $I = bh^3/12$                 | second moment of area of the cross-section                         | $x_1, x_2$                                         | displacements of the magnets from equilibrium (both positive to the right) |
| $\mathcal{M}(s), R$           | bending moment; radius of curvature                                | $u = x_2 - x_1, r = d + u$                         | change of separation; instantaneous separation                             |
| $k = 3EI/L^3$                 | tip stiffness of a spring                                          | $\gamma = -F'(d)$                                  | magnetic stiffness = $6\mu_0\mu_m^2/\pi d^5$                               |
| $\varphi(s)$                  | normalised deflection curve, eq. (9)                               | $\eta = x_1 + x_2, \xi = x_1 - x_2$                | normal coordinates                                                         |
| $M, \mu$                      | magnet mass; strip mass per unit length                            | $\omega_1, \omega_2; f_1, f_2$                     | angular / ordinary frequencies of the in-phase and anti-phase modes        |
| $m = M + \frac{33}{140}\mu L$ | effective oscillating mass                                         | $\eta_0, \xi_0$                                    | mode amplitudes, eq. (30)                                                  |
| $\mu_m, \mu_0$                | dipole moment of a magnet; permeability of free space              | $A, x_{20}$                                        | release displacement of spring 1; shift of spring 2 while spring 1 is held |
| $B(r), U(r), F(r)$            | on-axis field, energy, repulsive force                             | $C = 3\mu_0\mu_m^2/\pi^3 m$                        | constant in $f_2^2 - f_1^2 = Cd^{-5}$                                      |
|                               |                                                                    | $f_{\text{avg}}, f_{\text{beat}}, T_{\text{beat}}$ | $(f_1 + f_2)/2; f_2 - f_1; 1/f_{\text{beat}}$                              |